

RF Interference Suppression with Synthesis of Field-Programmable Microwave Photonic Filters Based on Hexagonal Meshes

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Abstract We demonstrate the synthesis of a field-programmable microwave photonic filter (MPF) on a 5×5 hexagonal mesh for RF interference suppression. Strong anti-interference ability of the synthesized integrated MPF architecture is validated with a 4 Gbps QPSK RF signal, achieving an EVM improvement of $>40\%$. ©2026 The Author(s)

Introduction

The proliferation of next-generation wireless devices has severely exacerbated electromagnetic interference (EMI) [1]. Microwave photonic filters (MPFs) provide a promising wideband, anti-EMI solution for radio frequency (RF) interference suppression [2]. For synthesizing versatile integrated MPFs (IMPFs) [3][4], field-programmable photonic integrated circuits (PICs) based on software-defined waveguide meshes have emerged [5][6][7]. However, configuring these meshes introduces a critical bottleneck: The topological search space grows exponentially with the scale of the PIC [8]. Manual configuration restricts designs to outmost waveguides, severely underutilizing internal resources, while empirical blind tuning of high-order IMPFs is inaccurate and time-consuming. Fully exploiting these platforms thus requires algorithms for comprehensive topological space exploration and precise parameter optimization. Consequently, without a systematic synthesis approach, the immense potential of large-scale programmable PICs for dynamic RF interference suppression remains a challenge.

Herein, we propose a field-programmable IMPF synthesis framework for RF interference suppression, experimentally validated on the 5×5 hexagonal PIC. By combining the IMPF architecture search with the analytical phase-voltage model, the system achieves frequency tuning

and precisely suppress RF interference. Furthermore, in a 4 Gbps QPSK RF link, the synthesized IMPF dynamically mitigates strong interference, significantly improving the error vector magnitude (EVM) from $\sim 60\%$ to $\sim 20\%$.

Operating Principle of Synthesizing IMPF

Figure 1 illustrates the synthesis process of the field-programmable IMPF for RF interference suppression. The framework identifies valid filter architectures under specific constraints and subsequently aligns the notch frequency with the target interference via voltage optimization.

This process consists of two major tasks. Firstly, for the PIC based on hexagonal waveguide meshes, when the constraints of the input optical port, the filter order and total optical path length are determined, obtain all the IMPF architectures that meet the conditions by using the proposed IMPF architecture search algorithm. Second, based on the IMPF's initial response and the interference frequency, the proposed IMPF voltage-mapping algorithm is utilized to optimize the voltages to achieve effective interference suppression.

As illustrated in Fig.2, a 5×5 hexagonal waveguide PIC proposed in our previous work [9] is adopted, the IMPF architecture search algorithm employs a graph-theoretic breadth-first search (BFS) framework with segmented routing logic to

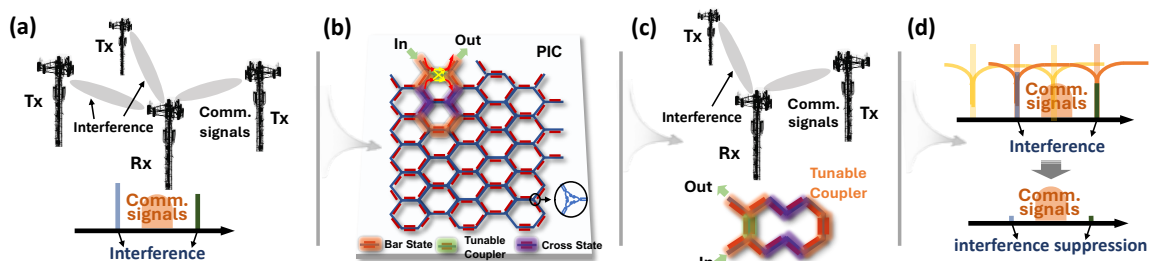


Fig. 1: Schematic diagram of the synthesis process for the proposed field-programmable microwave photonic filter.

systematically identify valid filter topologies within the mesh. Specifically, the synthesis framework abstracts the PIC into a graph-theoretic model, and the specific numbering method is shown in Fig. 2(a)-(d). The algorithm synthesizes valid topologies to meet specific architectural requirements by cascading four distinct segments. First, input access determines the shortest path to the initial TBU entry. Second, a micro-ring path search establishes the resonant cavity by identifying a closed mesh loop between a TBU's entrance and exit edges. Third, inter-stage bridging links the exit of the current TBU to the entry of the next, ensuring proper cascading order. Finally, terminal connection routes the signal from the last TBU exit to the nearest degree-2 boundary node, designating it as the egress port. Fig. 2(e) illustrates this architecture search for a second-order IMPF. During the search, defined pruning constraints maintain optical path correctness and physical validity within the expansive mesh, avoiding the inefficiencies of manual routing. Algorithm search time is governed by mesh scale (V vertices, E edges) and cascading order N . This segmented approach avoids exponential state-space explosion, ensuring the time complexity of the algorithm is $O(N \cdot M_{avg} \cdot (V+E))$, where physical constraints effectively suppress the candidate set M_{avg} .

Based on the initial spectral response of the synthesized IMPF architecture, the IMPF voltage-mapping algorithm optimizes the control voltages to align the notch frequency with the target interference. The algorithm employs constrained nonlinear optimization algorithm to identify the optimal voltages for precise frequency alignment and maximum notch depth. Based on an analytical phase-voltage model, an optimized voltage

range is then established around these values to effectively account for fabrication tolerances and environmental discrepancies, ensuring robust interference suppression.

Experimental Setup and Results

We constructed a RF signal transmission system to validate the interference suppression and signal of interest (SI) separation performance of first- and second-order IMPFs synthesized by the algorithm in Fig. 2(a) and 2(e).

A 10-dBm optical carrier at 193.4 THz, optimized via a polarization controller (PC), is fed into a Mach-Zehnder modulator (MZM). To simulate the SI, a 4 Gbps QPSK signal modulated at 13 GHz was generated by an arbitrary waveform generator (AWG) and amplified by an electronic amplifier (EA) to drive the MZM at its quadrature transmission point.

Concurrently, strong single- or double-frequency interference is simulated by coupling one or two additional tunable lasers into the link. Following power and polarization optimization, the combined signals are injected into the PIC, where the IMPF is precisely tuned to align its notch frequency with the interference. Post-PIC, the filtered signal is amplified by an erbium-doped fibre amplifier (EDFA) and converted back to the electrical domain via an avalanche photodiode (APD). A subsequent down-conversion is performed using an RF mixer driven by an 8-dBm, 18 GHz local oscillator (LO). As conceptualized in Fig. 3(a), without the IMPF, the interference would overlap with the SI band during down-conversion which is a challenge that conventional software filters fail to address. By effectively eliminating interference prior to the mixer, the synthesized IMPF prevents spectral aliasing, thereby substantially improving

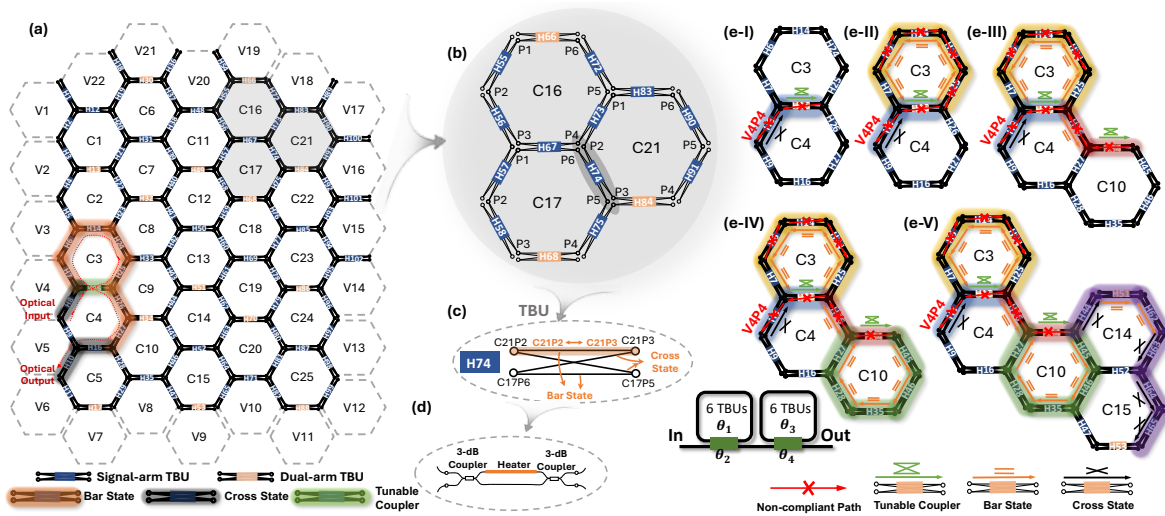


Fig. 2: (a)-(c) Numbering scheme for the physical and virtual hexagonal waveguide meshes. (d) Schematic of the TBU implemented by the MZI. (e) Synthetic second-order IMPF architecture example.

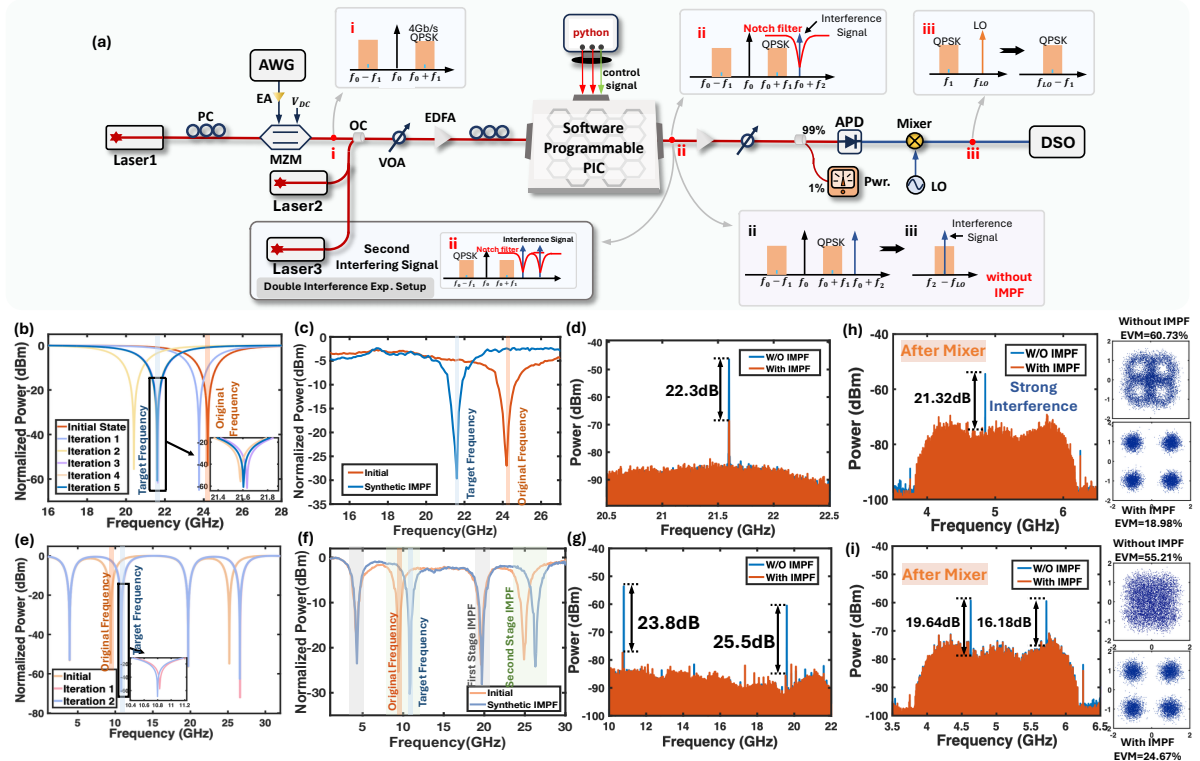


Fig. 3: (a) Experimental demonstration of the synthesized band-stop IMPF for suppressing interference signals in RF communication systems. (b)-(d) results for the first-order IMPF, (e)-(g) results for the second-order IMPF: (b) and (e) simulated convergence of the power spectrum during iterations, (c) and (f) measured electric spectrum before and after tuning; (d) and (g) spectrum comparison of interference before and after synthesizing the specified spectral response of IMPF.

the error vector magnitude (EVM) and overall communication quality.

For performance characterization, an RF source replaced the AWG and the mixer was bypassed to measure the response. In Fig. 3(b), The first-order IMPF notch converged from the initial 24.2 GHz to 21.6 GHz within five algorithm iterations. Meanwhile, the algorithm determined the voltage range that needed to be tuned for the two arms of the tunable coupler (TC). The voltage value of TC was controlled through python programming, and the response of IMPF was rapidly adjusted and aligned from the initial 24.2 GHz to 21.6 GHz as shown in Fig. 3(c), achieving a 22.3 dB suppression with <0.5% voltage deviation, as shown in Fig. 3(d). Second-order IMPF synthesis was also validated. As shown in Fig. 3(e), one notch was tuned from 9.4 GHz to 10.8 GHz while the 19.7 GHz notch remained fixed. As shown in Fig. 3(f), the actual second-order IMPF responses obtained via voltage programming before and after tuning match the simulated results and achieved robust dual-frequency suppression of 23.8 dB and 25.5 dB (Fig. 3(g)), proving the framework's efficacy in handling multi-frequency interference. The synthesized IMPF was applied to the 4 Gbps QPSK link. The first-order IMPF suppressed 21.32 dB of strong single-frequency interference, as shown in

Fig. 3(h). The EVM significantly improved from 60.73% to 18.98%, restoring clearly distinguishable constellation clusters. To address the more complex situation of dual-frequency interference, we synthesized a second-order IMPF, achieving suppression of 19.64 dB and 16.18 dB, respectively, as shown in Fig. 3(i). This architecture effectively reduced the EVM from 55.21% to 24.67%, demonstrating the robustness of the framework in multi-interference environments.

Conclusions

We demonstrated a synthesis framework of the programmable IMPF based on a 5×5 hexagonal mesh for interference suppression. The system achieved IMPF's notch frequency precise tuning with <0.5% error and >22.3 dB interference suppression. In a 4 Gbps QPSK link, the IMPF demonstrated a significant enhancement in signal quality, with improved EVM under strong interference. This work provides a key system-level verification for software-defined PICs, paving the way for future software-defined RF front-ends.

Acknowledgements

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